

A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics

Honour School of Economics and Management

Honour School of History and Economics

MICROECONOMICS

OUTLINE ANSWERS TO PROBLEMS

TRINITY TERM 2019

Wednesday 29 May, 09:30–12:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

*Answer **all** questions from Part A and **two** questions from Part B.*

Part A attracts 40% of the marks in total; each question in Part B attracts 30% of the marks.

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;

in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an economy with one consumer, one firm (owned by the consumer), and two types of good, X and Y .

The consumer owns the endowment of the economy, namely 14 units of good X , and her utility function is $u(x_c, y) = \ln x_c + 2 \ln y$.

The firm can transform good X into good Y ; if it uses an amount x_f of good X then it produces an amount $y = x_f^{2/3}$ of good Y .

- (a) Find the consumer's MRS and the firm's MRT.
- (b) Show that the allocation in which she consumes $x = 6$ and $y = 4$ is efficient.
- (c) Take good Y as the numeraire. What relative price of good X is required for this allocation to be a competitive equilibrium?
Find the firm's profit in this equilibrium, and verify that the consumer's budget constraint (which includes the firm's profit as her income) is satisfied.
- (d) Illustrate the equilibrium in a diagram showing the production possibility frontier of the economy.

[30%]

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1. For MRS & MRT, absolute values are also acceptable

(a) MRS: $\ln x_c + 2 \ln y = C \implies \frac{1}{x_c} + \frac{2}{y} y' = 0 \implies y' = \boxed{-y/2x_c}$.

MRT: PPF is $y = x_f^{2/3} = (E - x_c)^{2/3}$, $y^{3/2} = E - x_c \implies \frac{3}{2} y^{1/2} y' = -1 \implies y' = \boxed{-\frac{2}{3} y^{-1/2}}$.

- (b) When 6 units of good X are consumed, and the firm uses the remaining 8 to produce 4 units of good Y , the endowment of 14 is exhausted.

In this case, $MRS = -4/(2 \times 6) = -1/3$, $MRT = -\frac{2}{3} \times 4^{-1/2} = -1/3$.

- (c) The price ratio must equal $-MRS$ so $p_x/p_y = 1/3$, say $p_x = 1/3, p_y = 1$.

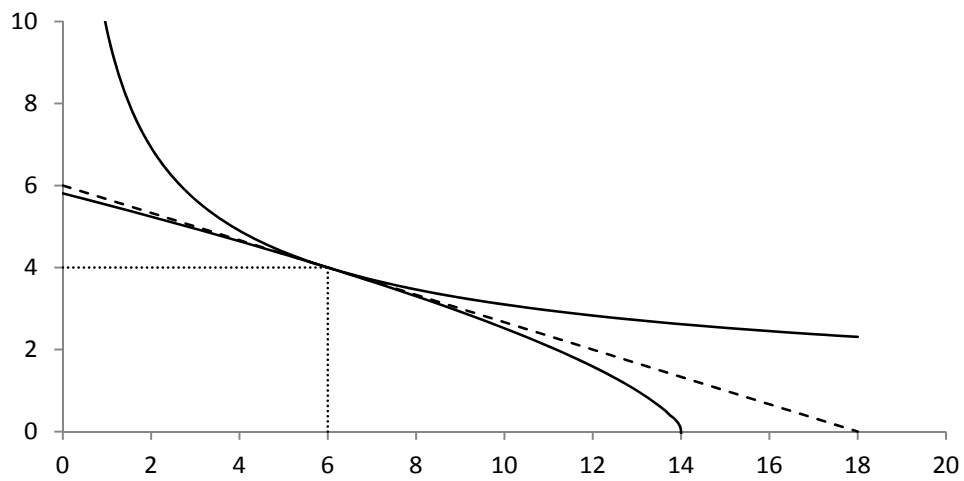
The firm receives 4 from selling 4 units of good Y but pays $8/3 = 2\frac{2}{3}$ for the 8 units of good X as input, resulting in a profit of $1\frac{1}{3}$.

The consumer's expenditure on the two goods is $6p_x + 4p_y = \boxed{6}$.

Her income is the value of her endowment, $14p_x = 4\frac{2}{3}$, plus the firm's profit, $1\frac{1}{3}$, which also comes to 6.

So her budget constraint is indeed satisfied.

(d) PPF, Indifference curve, and separating price line



2. (a) What does it mean for a strategy to be dominant?

If players choose dominant strategies, will they play according to a Nash Equilibrium?

And if they play according to a Nash Equilibrium, will they play dominant strategies?

- (b) For each of the following two-player one-shot games, discuss whether or not you can predict how the players will play.

Game 1. Each player decides between *Green* and *Red* and they simultaneously announce their choices, which result in the following pay-offs:

if they chose the same action then they each receive 5, otherwise the one that chose *Green* receives 6 and the one that chose *Red* receives 4.

Game 2. Each player decides between *Blue* and *Yellow* and they simultaneously announce their choices, which result in the following pay-offs:

if they both chose *Blue* then they each receive 3, whereas if they both chose *Yellow* then they each receive 5; otherwise the one that chose *Blue* receives 4 and the one that chose *Yellow* receives 2.

- (c) Comment on the usefulness of the Nash Equilibrium concept for predicting the outcome of a game.

[20%]

2. (a) A strategy is (weakly) dominant if it is a best response to any strategy of the opponent.

If players choose (weakly) dominant strategies then they are all playing best responses and hence a Nash equilibrium (e.g. prisoner's dilemma).

However, there are Nash equilibria in which players do not play (weakly) dominant strategies (e.g. coordination games, matching pennies).

- (b) *Game 1.* Each player has a dominant strategy to choose *Green*. The prediction is that both choose *Green* (assuming rationality).

Game 2. Both (*Blue*, *Blue*) and (*Yellow*, *Yellow*) are Nash equilibria.

(There is also a mixed-strategy Nash equilibrium – candidates do not need to calculate the mixing probabilities of $(\frac{1}{2}, \frac{1}{2})$ in order to answer the question.)

The multiplicity of equilibria makes it difficult to predict behaviour.

- (c) Nash equilibrium provides a good prediction if a game has a unique equilibrium. Otherwise, it is less clear that players will play according to one particular Nash equilibrium, unless one of them can be selected using focal points, Pareto dominance, risk dominance, etc.

In *Game 2*, (*Yellow*, *Yellow*) is Pareto dominant; neither is risk dominant.

3. (a) Define the certainty equivalent of a lottery. How can it be used to determine whether or not someone should participate in a lottery?

There is a lottery whereby anyone with a ticket will win either £12 or nothing, each with probability $\frac{1}{2}$.

- (b) Janet is an expected-utility maximiser with utility function $u(y) = 2y^{1/2}$ over final wealth y . She has initial wealth of £4 and also a ticket for the lottery.

What is her expected utility?

What is the lowest price that she would be willing to sell the ticket for?

- (c) Sam is also an expected-utility maximiser and has the same utility function as Janet. Sam also has initial wealth of £4 but doesn't have a ticket for the lottery.

What is the highest price that Sam would be willing to pay for a ticket?

- (d) Illustrate your results from part (b) and part (c) in a diagram showing wealth after winning the lottery on the horizontal axis, and wealth after no pay-out from the lottery on the vertical axis.

Which is larger, Janet's lowest selling price or Sam's highest buying price, or are they the same? Explain.

[30%]

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3. (a) The certainty equivalent is y^c where $u(y^c) = \bar{u}$, where $\bar{u}$ is the expected utility from taking part in the lottery.

An EU maximiser would take part in a lottery if and only if the certainty equivalent is at least as large as final wealth from not taking part.

- (b) Expected utility $= \frac{1}{2}u(4 + 12) + \frac{1}{2}u(4 + 0) = \sqrt{16} + \sqrt{4} = 4 + 2 = 6$.

Indifferent selling ticket at price p_s such that $u(4 + p_s) = 6$,

i.e. $2\sqrt{4 + p_s} = 6$, $4 + p_s = 9$, $\boxed{p_s = 5}$.

- (c) Indifferent between buying ticket at price p_b and not buying:

$$\frac{1}{2}u(4 + 12 - p_b) + \frac{1}{2}u(4 + 0 - p_b) = u(4), \quad \sqrt{16 - p_b} + \sqrt{4 - p_b} = 2\sqrt{4}.$$

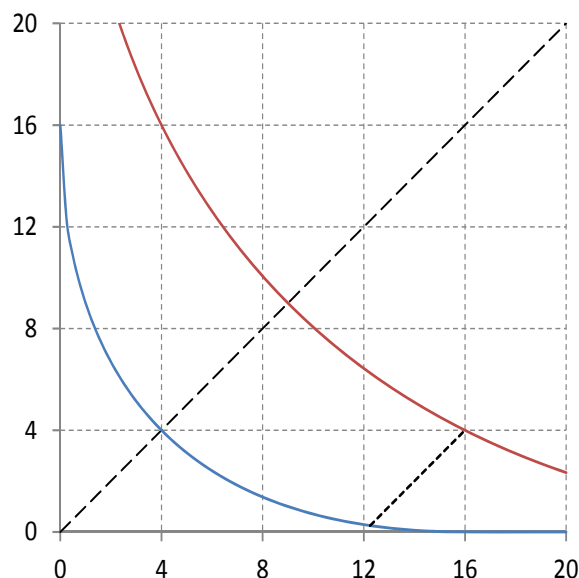
Square both sides: $16 - p_b + 4 - p_b + 2\sqrt{(16 - p_b)(4 - p_b)} = 16$;

simplify & rearrange: $\sqrt{(16 - p_b)(4 - p_b)} = 2 - p_b$;

square again & simplify: $64 - 20p_b = 4 - 4p_b$, $\boxed{p_b = 15/4 = 3\frac{3}{4}}$.

- (d) Janet starts on the $\bar{u} = 6$ indifference curve at $(16, 4)$ and if she simply gave her ticket away she'd be on the $\bar{u} = 4$ indifference curve at $(4, 4)$. To compensate, we have to get her back onto the original indifference curve at $(9, 9)$.

Sam starts on the $\bar{u} = 4$ indifference curve at $(4, 4)$ and if simply given a ticket would be on the $\bar{u} = 4$ indifference curve at $(16, 4)$. The point on the original indifference reached by deducting the same amount in each state of the world is $(12\frac{1}{4}, \frac{1}{4})$.



Clearly $p_s > p_b$. They would be the same if the utility function exhibited Constant Absolute Risk Aversion, but their utility captures Constant Relative Risk Aversion, and hence Decreasing Absolute Risk Aversion; as a consequence, the distance between two indifference curves (as measured along a 45° line) decreases as the lottery gets further away from the certainty line.

4. A risk-neutral principal hires a risk-averse agent to undertake a project. There are two possible effort levels that the agent can exert, $e = 0$ or 1 , and two possible outcomes for the gross profit to the principal, $\pi_L < \pi_H$. If $e = 0$, the probabilities of π_L, π_H are $5/6, 1/6$ respectively; if $e = 1$, the probabilities of π_L, π_H are $5/8, 3/8$ respectively.

The agent's utility from receiving a wage w and exerting effort e is $\sqrt{w} - 5e$, and the agent has a reservation utility of 10 .

- (a) When effort is observable (& contractible), what contract should the principal offer the agent if she wants high effort, and why?

Now assume that effort is unobservable: in the contract between the two parties, w cannot depend on e directly, but can depend on the outcome, π_k . Thus a contract will be of the form $\{w_L, w_H\}$, where w_k is the wage paid by the principal to the agent if the gross profit is π_k . Assume that the principal wants to induce high effort.

- (b) If the lower wage, w_L , is 36 , find the higher wage, w_H , such that the participation constraint is binding.

Show that, with these values of w_L and w_H , the contract is incentive compatible.

- (c) In words, what is the "agency cost" associated with unobservable effort?

Calculate the expected wage, and hence determine the agency cost in this case.

[20%]

4. (a) *Observable effort ...*

Wage is fixed and just compensates for high effort, $\sqrt{w^*} - 5 = 10$, so $\boxed{w^* = 225}$.

- (b) *Unobservable effort; principal wants to induce high effort ...*

Write v_L and v_H for the utility value of w_L and w_H .

Participation constraint: $\frac{5}{8}v_L + \frac{3}{8}v_H - 5 \geq \bar{u}$, i.e. $\frac{5}{8}v_L + \frac{3}{8}v_H - 5 = 10$.

$w_L = 36 \implies v_L = 6$, so Participation constraint is $\frac{5}{8} \times 6 + \frac{3}{8}v_H = 15$, equivalently $30 + 3v_H = 120$, giving $v_H = 30$ and therefore $\boxed{w_H = 900}$.

Incentive constraint is $\frac{5}{8}v_L + \frac{3}{8}v_H - 5 \geq \frac{5}{6}v_L + \frac{1}{6}v_H - 0$.

Substitute $v_L = 6, v_H = 30$: LHS is $10 (= \bar{u})$; RHS is also 10 .

So Incentive constraint is satisfied (as an equality).

- (c) Agency cost is difference in the expected wage when effort is unobservable and the fixed wage when effort is observable (given high effort), in each case to make sure the agent reaches his reservation utility.

As such, it is the risk premium associated with the lottery faced by an agent who exerts high effort.

$$E[w] = \frac{5}{8}w_L + \frac{3}{8}w_H = 360.$$

Agency cost is thus $360 - 225 = \boxed{135}$.

Part B

10. Consider an exchange economy with three sorts of good: apples (A), bananas (B), and cherries (C).

There are 20 traders, 10 of type 1, and 5 each of types 2 and 3, whose initial allocation is as follows:

- type 1 traders each own 10 apples,
- type 2 traders each own 5 bananas,
- type 3 traders each own 20 cherries.

All traders have the same utility function that depends only on their consumption of apples (x_A), bananas (x_B), and cherries (x_C):

$$u(x_A, x_B, x_C) = x_A^{1/2} x_B^{1/4} x_C^{1/4}.$$

Let p_A , p_B , and p_C denote the prices of the three sorts of good, and normalise p_C (the price of cherries) to 1. Suppose that all traders can trade freely taking prices as given.

- (a) Will all types of trader engage in trade? Why?

What is the demand for apples, bananas and cherries for each type of trader?

[30%]

- (b) Calculate the excess demand for apples and bananas in the economy.

Calculate the competitive equilibrium prices.

What is the excess demand for cherries in this competitive equilibrium? Why?

[40%]

Now consider a different allocation (apples, bananas, cherries) for the traders:

- each type 1 trader has $(9, 9/4, 9)$,
- each type 2 trader has $(1, 1/4, 1)$,
- each type 3 trader has $(1, 1/4, 1)$.

- (c) Is this allocation Pareto efficient?

If it is, find the supporting competitive equilibrium prices.

If it isn't, propose a Pareto-improving trade.

[30%]

10. (a) Each agent is endowed with only one sort of good, but the MRS between any two sorts of good tends to infinity as the consumption of one of the goods tends to zero. Therefore, all agents will want to trade.

It is not enough to say that each agent's utility depends on each good or that if the agent consumes zero of some good then he gets zero utility – cardinal utilities are not meaningful here so the argument must rely on marginal utilities/MRS's.

It is also not enough to simply say that, in the equivalent problem where $v = \ln u$, utility would be undefined at zero.

- (a) Interior solutions for the above reasons.

For each type, the share of the budget spent on a good is the weight placed on that good in the utility function (i.e. the exponent $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$).

Recall $\boxed{p_C = 1}$.

$$\begin{aligned} x_A^1(p) &= \frac{\frac{1}{2} \times 10 p_A}{p_A} = 5, & x_B^1(p) &= \frac{\frac{1}{4} \times 10 p_A}{p_B} = \frac{5 p_A}{2 p_B}, & x_C^1(p) &= \frac{\frac{1}{4} \times 10 p_A}{p_C} = \frac{5 p_A}{2} \\ x_A^2(p) &= \frac{\frac{1}{2} \times 5 p_B}{p_A} = \frac{5 p_B}{2 p_A}, & x_B^2(p) &= \frac{\frac{1}{4} \times 5 p_B}{p_B} = \frac{5}{4}, & x_C^2(p) &= \frac{\frac{1}{4} \times 5 p_B}{p_C} = \frac{5 p_B}{4} \\ x_A^3(p) &= \frac{\frac{1}{2} \times 20 p_C}{p_A} = \frac{10}{p_A}, & x_B^3(p) &= \frac{\frac{1}{4} \times 20 p_C}{p_B} = \frac{5}{p_B}, & x_C^3(p) &= \frac{\frac{1}{4} \times 20 p_C}{p_C} = 5 \end{aligned}$$

- (b) Total endowments: $e_A = 10 \times 10 = 100$, $e_B = 5 \times 5 = 25$, $e_C = 5 \times 20 = 100$.
Excess demand $z(p) = x(p) - e$:

$$\begin{aligned} z_A(p) &= 10 \times 5 + 5 \times \frac{5 p_B}{2 p_A} + 5 \times \frac{10}{p_A} - 100 = \frac{25 p_B}{2 p_A} + \frac{50}{p_A} - 50 \\ z_B(p) &= 10 \times \frac{5 p_A}{2 p_B} + 5 \times \frac{5}{4} + 5 \times \frac{5}{p_B} - 25 = \frac{25 p_A}{p_B} + \frac{25}{p_B} - \frac{75}{4} \end{aligned}$$

- (b) Set excess demands to zero:

$$\begin{aligned} z_A(p) = 0 &\implies 25 p_B + 100 - 100 p_A = 0 \\ z_B(p) = 0 &\implies 100 p_A + 100 - 75 p_B = 0 \end{aligned}$$

$$\text{Add: } 50 p_B = 200, \quad \boxed{p_B = 4}, \quad \text{leading to } \boxed{p_A = 2}$$

- (b) From Walras Law, excess demand for cherries must also be zero.

$$\text{Check: } z_C(2, 4, 1) = 10 \times \frac{5 p_A}{2} + 5 \times \frac{5 p_B}{4} + 5 \times 5 - 100 = 50 + 25 + 25 - 100 = 0$$

(c) For MRS, absolute values are also acceptable

For each type of trader: $u_A = \frac{1}{2}x_A^{-1}u$, $u_B = \frac{1}{4}x_B^{-1}u$, $u_C = \frac{1}{4}x_C^{-1}u$, so

$$MRS_{AB} = -\frac{2x_B}{x_A}, \quad MRS_{AC} = -\frac{2x_C}{x_A}, \quad MRS_{BC} = -\frac{x_C}{x_B}$$

For Pareto efficiency of the allocation, we require that the MRS between any two particular sorts of good is the same for each type of trader:

$$\begin{aligned} MRS_{AB}^1 &= -\frac{2 \times 9/4}{9}, & MRS_{AB}^2 &= -\frac{2 \times 1/4}{1}, & MRS_{AB}^3 &= -\frac{2 \times 1/4}{1}, & MRS_{AB} &= -1/2 \\ MRS_{AC}^1 &= -\frac{2 \times 9}{9}, & MRS_{AC}^2 &= -\frac{2 \times 1}{1}, & MRS_{AC}^3 &= -\frac{2 \times 1}{1}, & MRS_{AC} &= -2 \\ MRS_{BC}^1 &= -\frac{9}{9/4}, & MRS_{BC}^2 &= -\frac{1}{1/4}, & MRS_{BC}^3 &= -\frac{1}{1/4}, & MRS_{BC} &= -4 \end{aligned}$$

For Pareto efficiency, we also require that the new allocation exhausts the endowments:

$$\begin{aligned} A: & 10 \times 9 + 5 \times 1 + 5 \times 5 = 100 = e_A \\ B: & 10 \times 9/4 + 5 \times 1/4 + 5 \times 1/4 = 25 = e_B \\ C: & 10 \times 9 + 5 \times 1 + 5 \times 5 = 100 = e_C \end{aligned}$$

So the outcome is Pareto efficient.

The MRS's between any two goods are the same for each type of trader and the endowments are exhausted.

Therefore, the allocation is on the contract curve.

Hence, by the Second Welfare Theorem, there is a price vector that supports the allocation as an equilibrium.

The price ratios are equal to the (absolute value of the) MRS's – and are the same as for the previous equilibrium, namely (2, 4, 1).